IEEE SSCS®

Solid-State Circuits Society

Task 5 - Level 1

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Regression Assignment:

Problem Statement:

Use the California Housing Prices dataset (from the 1990 California census) to build and evaluate regression models for predicting Median House Value.

Dataset Features:

- **Median House Value**: Median house value within a block (USD) [$]
- **Median Income**: Median household income within a block (in tens of thousands of USD) [10k$]
- **Median Age**: Median age of houses within a block (years)
- **Total Rooms**: Total number of rooms within a block
- **Total Bedrooms**: Total number of bedrooms within a block
- **Population**: Total number of residents within a block
- **Households**: Total number of households within a block
- **Latitude**: Geographic latitude (°)
- **Longitude**: Geographic longitude (°)
- **Distance to Coast**: Distance to nearest coastal point (m)

- **Distance to Los Angeles**: Distance to city center (m)
- **Distance to San Diego**: Distance to city center (m)
- **Distance to San Jose**: Distance to city center (m)
- **Distance to San Francisco**: Distance to city center (m)

Required Tasks

- **Split** the dataset randomly into:
  - **70% Training**
  - **15% Validation**
  - **15% Testing**

    (Do not use the test set while tuning model parameters.)

- **Implement Linear Regression from scratch**:

  1. Compute weights using the Normal Equation (closed-form solution): $$w = (X^T X)^{-1} X^T t_Y$$
     - This provides the exact optimal solution without iteration.

  2. Gradient Descent :
     - Initialize weights **W** (zeros or random).
     - Choose a learning rate α.

- Define the cost function (MSE) then Compute gradients
- And then use the Update rule:

$$w := w - \alpha \nabla J(w)$$

- Repeat until convergence or max iterations.

  Note: Feature scaling is recommended as it helps achieve faster convergence.

- **Compare and comment on both approaches (closed-form solution Vs Gradient Descent) in a Markdown.**

- **Repeat**

  - **Linear** Regression

using **Scikit-learn** implementations and comparing them with your from-scratch results.

- Report on the following metrics for all models:
  - **Mean Squared Error (MSE)**
  - **Mean Absolute Error (MAE)**

(You must **implement both metrics from scratch** using basic numerical operations (e.g., NumPy))

- Add clear comments on results and provide a comparison between all models.